# Arslan Arshad

Senior AI & Backend Engineer · GenAI · LLM Systems · Full-Stack

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*Fully Remote · US Eastern / EU Central overlap (5+ hrs/day) · Immediate availability*

## Professional Summary

Senior AI & Backend Engineer with 4 years of experience building production-grade agentic systems and high-throughput backend infrastructure. Proven track record of owning the full GenAI lifecycle – from RAG architecture and LangGraph orchestration to backend optimization, evaluation, and production deployment. Reduced a 4-hour manual research workflow to under 10 minutes and delivered 35–40% API performance improvements across production systems. Expert in bridging the gap between AI prototypes and scalable, business-critical software.

## Technical Skills

**AI & GenAI:** LangChain, LangGraph, RAG Pipelines, LLM Orchestration, Prompt Engineering, Agentic AI, Multi-Agent Systems, Semantic Search, Embeddings, Vector Databases (Pinecone, FAISS), LLaMA, Gemini, Groq API

**Backend:** FastAPI, Django, Django REST Framework, Flask, Celery, Django Channels (WebSockets), Microservices, REST API Design, Event-Driven Architecture, Multi-Tenant Systems

**Frontend:** ReactJS, HTML5, CSS3, Material UI (MUI)

**Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQLite

**Cloud & DevOps:** AWS (EC2, RDS, S3), Docker, CI/CD, Cloud-Native Deployment

**Auth & Integrations:** JWT, OAuth2, Stripe, Plaid, Twilio, OpenAI-compatible APIs

**Languages:** Python, JavaScript, Java

**Tools & Process:** Git, GitHub, Postman, Jira, Agile/Scrum, Async Remote Collaboration, Technical Documentation

## Professional Experience

**Senior Software Engineer, AI & Backend | Kinetic Intelligence** *Remote (Dubai, UAE) · Feb 2026 – Present*

- Reduced daily analyst workload by several hours by shipping production-grade **GenAI case analysis workflows** and automated prompt engineering pipelines for investigative operations.
- Achieved a **40% reduction in API response times** for complex investigative queries by combining Redis caching, PostgreSQL indexing, and query-level performance optimization.
- Architected high-throughput Django and PostgreSQL REST APIs with **sub-100ms response targets** for latency-sensitive investigative data workloads.
- Owned full-stack feature delivery across Django and ReactJS, taking features from system design and implementation through production release.

**Full-Stack Software Engineer | Inovaqo** *Lahore, Pakistan · Jan 2025 – Jan 2026*

- Reduced a **4-hour manual market research workflow to under 10 minutes** by architecting a multi-agent LangGraph system for planning, research, semantic retrieval, tool use, and structured report generation.
- Improved API response times by **35%** through query profiling, selective database indexing, and layered Redis caching across high-traffic endpoints.
- Replaced brittle keyword lookups with a **RAG-based semantic search pipeline** using LangChain and Pinecone for intelligent retrieval across large document corpora.
- Architected a **multi-tenant SaaS backend** using Django REST Framework and FastAPI with isolated customer environments, independent data pipelines, and role-based access control.
- Integrated Stripe, Plaid, and Twilio into production APIs and implemented JWT and Google OAuth2 authentication for secure user access.

- Owned end-to-end delivery of AI and backend features from architectural design through Dockerized AWS deployment, bridging complex LLM workflows with scalable SaaS infrastructure.

**Associate Software Engineer | CodingCops** *Lahore, Pakistan · Dec 2022 – Jan 2025*

- Engineered and shipped full-stack production systems using Django REST Framework and ReactJS, owning API design, database architecture, and frontend integration across multiple client engagements.
- Designed normalized PostgreSQL and MySQL schemas and scalable REST API contracts, establishing maintainable foundations for evolving business requirements.
- Improved maintainability by standardizing API response structures, frontend integration patterns, and reusable ReactJS components across projects.

## Key Projects

**AI-Powered Market Research Platform** *FastAPI | React | LangChain | LangGraph | Pinecone | Groq API | LLaMA*

- Designed a **multi-agent AI system** using LangGraph that autonomously plans, researches, and generates structured market analysis reports, reducing a **4-hour manual workflow to under 10 minutes**.
- Implemented a production RAG pipeline using Pinecone and embeddings to ground LLaMA responses in retrieved business data and improve factual consistency.
- Added structured output validation and LLM-based evaluation workflows to assess generated reports for relevance, completeness, and answer quality.
- Built a modular FastAPI backend with Pydantic validation, clean agent-service separation, and REST APIs consumed by a React frontend.

**YouTube Talker – AI Video Content Analyst** *Flask | LangChain | Gemini | FAISS | YouTube API*

- Built an end-to-end GenAI pipeline that extracted YouTube transcripts, generated embeddings, indexed content with FAISS, and enabled natural-language querying over long-form video content.
- Integrated Google Gemini for context-aware answer generation over retrieved transcript chunks, improving response grounding for long-form content.

**InsightRAG – Self-Grading Document Q&A Agent** *FastAPI | LangGraph | LangChain | FAISS | Groq*

- Built a self-correcting **RAG agent** with LangGraph that rewrites queries, grades retrieved context, and automatically refines retrieval when evidence is insufficient instead of answering blindly.
- Designed a **multi-signal confidence scoring model** combining vector similarity, lexical overlap, and LLM-based relevance grading to route low-confidence answers to explicit fallbacks and reduce hallucination risk.

## Education

**BS Software Engineering** *2019 – 2023*
Punjab University College of Information and Technology (PUCIT), Lahore, Pakistan

## Certifications

- **Mastering RAG: Build Smart, Data-Driven Applications** – IBM Skills Network (2026)
- **Generative AI with Vertex AI: Prompt Design** – Google Cloud Skills Boost

*Open to fully remote Software Engineer, Senior Software Engineer, AI Engineer, Backend Engineer, and GenAI Engineer opportunities across US, EU, UK, Middle East, and global teams. Experienced in async-first collaboration, cross-functional communication, technical documentation, and overlapping with US Eastern and EU Central time zones. Immediate availability for full-time remote roles.*